

Website Link: <http://classwork.engr.oregonstate.edu:8000/Avidex>

Executive Summary:

For our Project Step 1 draft, our database only modeled Birds, Nests, Sightings, and Rarities. In the reviews, it was pointed out that for a personal collection app, there needed to be an entity for a person whose collection it is, or else there's no way to say whose Avidex a sighting belonged to. That observation drove our largest structural change: we added Birders, then Rewards and BirdsList to track species totals. Two of our three M:N relationships did not exist in the original proposal. The reviews also pointed out many small problems, helping us fix mismatched data types, a missing PK, CamelCase naming issues, and flaws in our overview.

For Project Step 2, we received feedback that our ERD and Schema diagrams were illegible, so we revised them to be cleaner and easier to read. The next point of feedback we received concerned our composite primary keys. We didn't realize that we had to define both keys as primary keys to create a composite key, and we had to go back into our DDL.sql to properly declare them. We also had to add ON DELETE CASCADE and ON UPDATE CASCADE to our dependent entities to resolve update and delete foreign key checks. We also removed the requirement of a database named `Avidex` to be the parent of our tables in order for our DDL.sql to properly run. After that, we fixed some discrepancies between our PDF and DDL.sql, like fixing some typos and changing Sighting.gpsLocation and Sighting.time to allow NULL values.

Many of our reviewers in Project Steps 3 & 4 noted that our UI was difficult to use; for instance, the Create and Update sections were not organized. Based on that feedback, we decided to display the fields vertically. Reviewers also pointed out that most of our CRUD operations were not functioning, which we have since implemented correctly.

Title Page

Avidex Project

A Gamified Birding Collection App

A Formal Proposal Prepared for Ms. Visor Super
640A Kerr Administration Building, Corvallis, OR 97331

August 13, 2026
CS340 Section 400

Created By	
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Overview

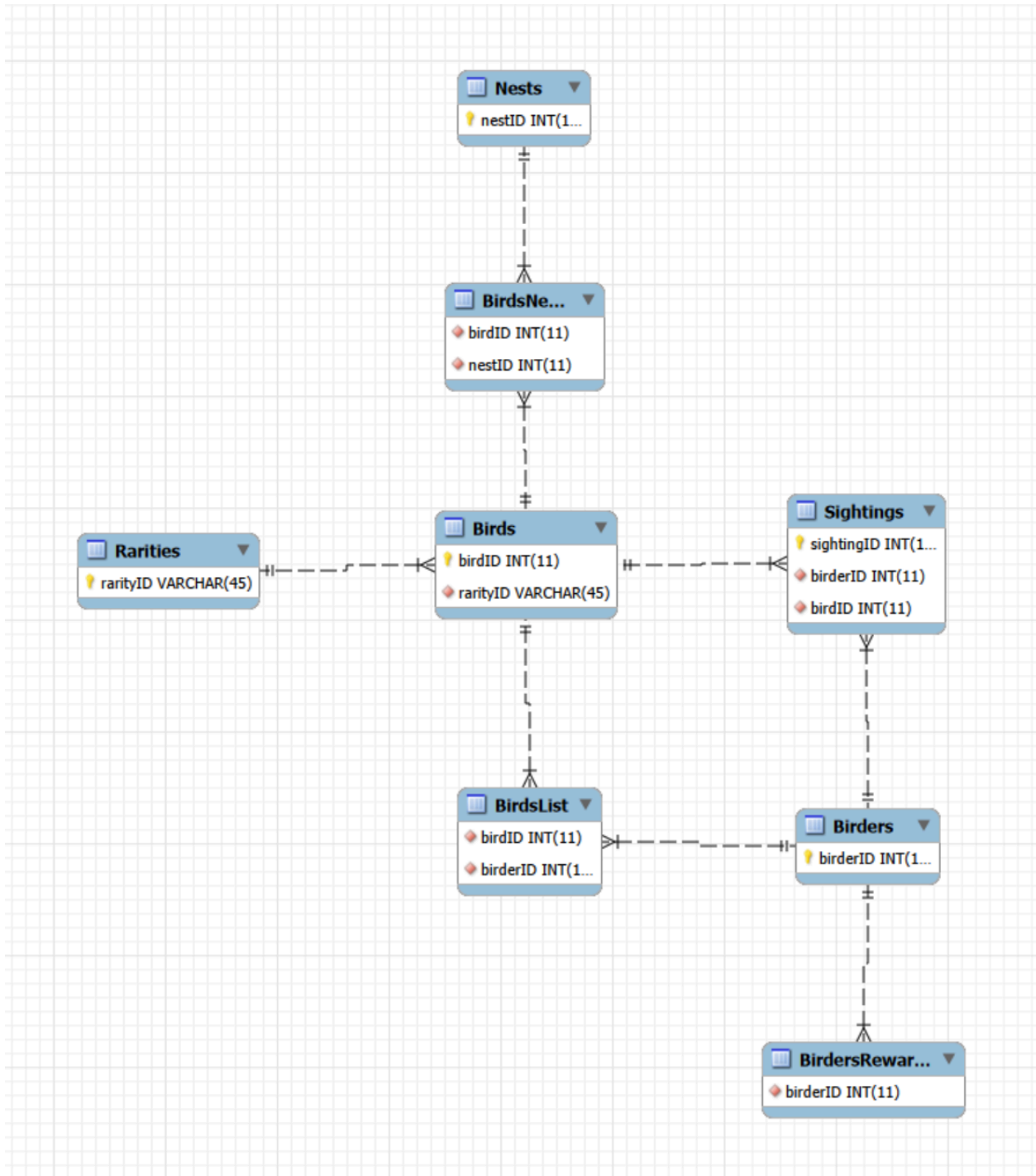
The Avidex Project team requests that Ms. Visor Super review the following Formal Proposal and approve it for further development. Specifically, the Avidex Project team requests that a gamified birding application be developed. Despite the strong interest in birding, we feel there is a severe lack of applications for tracking a birder's sightings in a fun and unique way. Therefore, we expect to create a birdwatching application meant for birding enthusiasts and newcomers. We're catering to a demographic of about 96 million birders in the U.S. — about $\frac{1}{3}$ of the population — and hope to help at least 20,000 birders keep track of 2,000 sightings each! With a Pokédex-like UI and setup, our application will encourage users to go out into the world and look for birds in nature to earn incremental rewards. Soon enough, users will go out into the world and "collect" up to 1,107 birds within their Avidex.

Database Outline

- **Birds-** records the details of Bird species that will populate the Avidex
 - birdID: int, PK, AI, UQ, not NULL — An integer to keep track of the specific bird's ID
 - commonName: varchar, not NULL — The bird species' common name i.e. American Crow, Blue Jay
 - species: varchar, UQ, not NULL — The specific bird's species name i.e. Corvus Corax, Cyanocitta cristata
 - callUrl: varchar, NULL — A URL to link to the call of the bird
 - wingspan: varchar, not NULL — A short description of the range of the wingspan for the bird
 - size: varchar, not NULL — A short description of the size of the bird
 - identifyingMarks: text, not NULL — A description of the identifying features of the bird
 - range: text, not NULL — A description of the bird's range
 - description: text, not NULL — Any interesting information about the bird's lifestyle and characteristics
 - photographUrl: varchar, not NULL — A URL to link to image(s) of the bird
 - matingSeason: varchar, not NULL — A description of the mating season of the bird
 - rarityID: varchar, FK, not NULL — The rarity of the bird
 - Relationships:
 - 1:M relationship to Sightings with a FK of Sightings.birdID
 - 1:M relationship to BirdsNest with a FK of BirdsNests.birdID
 - M:1 relationship to Rarity with a FK of Birds.rarityID
- **Rarities-** categorical table that records the rarity of each Bird
 - rarityID: varchar, PK, UQ, not NULL — The rarity of the bird in our app, i.e., Common, Uncommon, Rare, Legendary
 - Relationships:
 - 1:M relationship to Birds
- **BirdsNests-** composite entity for the M:N relationship of Birds to Nests
 - nestID: int, FK, not NULL — A foreign key for the Nests list
 - birdID: int, FK, not NULL — A foreign key for the Birds list
 - Relationships:
 - M:1 relationship to Nest with a FK of BirdsNests.nestID
 - M:1 relationship to Birds with a FK of BirdsNests.birdID
- **Nests-** records the details of various nests that birds build
 - nestID: int, AI, PK, UQ, not NULL — An ID for this type of nest
 - type: varchar, not NULL — The type of nest this bird has, i.e., Scrape, Cup, Burrow
 - location: varchar, not NULL — The location of this type of nest, i.e., Cliff, Brush, Tree
 - Relationships:
 - 1:M relationship to BirdsNests with a FK of BirdsNests.nestID
- **Sightings-** a transaction table to record the individual bird sightings
 - sightingID: int, PK, AI, UQ, not NULL — The ID for the sighting
 - birderID: int, FK, not NULL — The foreign key to link back to the Birder who made the sighting
 - birdID: int, FK, not NULL — The foreign key for the bird that was sighted
 - birdCount: int — The # of the specific bird they saw
 - gpsLocation: point, NULL — The location they saw it at
 - time: datetime, NULL — The time and day the bird was spotted
 - Relationships:
 - M:1 relationship with Birds with an FK of Sightings.birdID
 - M:1 relationship with Birders with an FK of Sightings.birderID
- **BirdsList-** tracks the list of birds a user has seen and the total count
 - birderID: int, FK — the ID for the users
 - birdID: int, FK — the bird they saw
 - count: int NOT NULL — The number of times they saw the specified bird
 - Relationships:
 - 1:M relationship with Birders with FK of BirdsList.birderID

- 1:M relationship with Birds with a FK of BirdsList.birdID
- **Birders-** user entity
 - birderID: int, PK, AI, UQ, NOT NULL — The id of the birder
 - birderName: varchar, UQ, NOT NULL — The name of the birder
 - points: int — The amount of points this birder has
 - Relationships:
 - 1:M relationship with BirdsList with a FK of BirdsList.birderID
 - 1:M relationship with BirdersRewards a FK of BirdersRewards.birderID
- **BirdersRewards-** composite entity for the M:N relationship of Birder to Rewards
 - birderID: int, FK, not NULL — The ID of the birder
 - rewardID: int, FK, not NULL — The ID of the reward
 - Relationships:
 - M:1 relationship to Birders with a FK of BirdersRewards.birderID
 - M:1 relationship to Rewards with a FK of BirdersRewards.rewardID
- **Rewards-** categorical entity for types of rewards
 - rewardID: int, PK, AI, UQ, NOT NULL — The ID of this reward
 - name: varchar, UQ, NOT NULL — The name for the reward
 - description: varchar, NOT NULL — A description of the reward
 - threshold: int, NOT NULL — The minimum amount of points needed to earn the reward
 - Relationships: 1:M with BirdersRewards with a FK BirdersRewards.rewardID

ER Diagram:



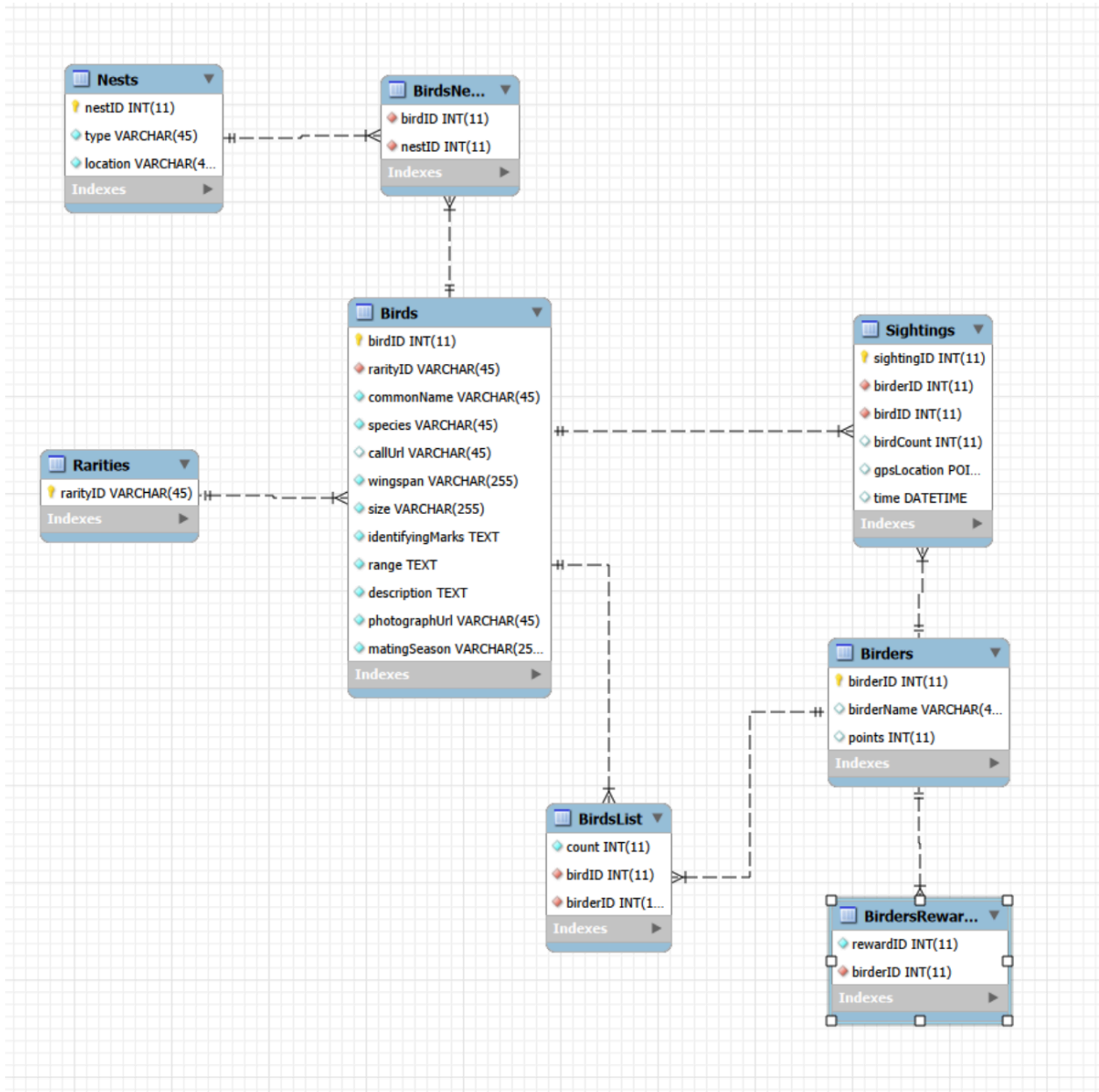
Schema Design and Normalization:

We started by writing out a sample sighting data table, with one row per observation and columns for the birder, the bird, its rarity, its nest styles, and the observation details. The report violated first normal form in one place. Because a species may build more than one style of nest, the nest column held a repeating group. The Bald Eagle row listed both a platform nest in a tree and a platform nest on a cliff in a single cell. We removed the repeating group by extracting nests into their own entity and joining it to Birds through the BirdsNests intersection table, which stores one row per bird/nest pairing. Every attribute now holds a single value and every table has a defined primary key, so the design is in 1NF.

For second normal form we looked at our three intersection tables. BirdsNests and BirdersRewards have no non key attributes, making partial dependency impossible. An earlier draft carried a description attribute on BirdsNests so we removed it because it described the nest style rather than the pairing, which made it partially dependent on nestID alone. That information now lives in Nests. BirdsList has one non-key attribute, count, which is meaningless without both halves of its key since it records how many times this birder saw this bird. Every other table uses a single-attribute primary key. The design is therefore in 2NF.

We then looked for transitive dependencies. Rarity was repeated on every row of the report and is now a separate entity keyed on the label itself, with Birds carrying only the foreign key, so each tier is stored once. Nest type and location describe the nest rather than the sighting and are stored once per nest record. Bird attributes such as wingspan, range, and description depend on the bird rather than the sighting event and are stored once in Birds.

Schema Diagram



Tables

Table: Rarities

rarityID
Common
Uncommon
Rare
Legendary

Table: Birds (part 1 of 3 — identity and media)

birdID	commonName	species	rarityID	callUrl	photographUrl
1	Bald Eagle	Haliaeetus leucocephalus	Uncommon	https://baldeaglecallhere.com	https://baldeaglephotoshere.com
2	American Robin	Turdus migratorius	Common	https://americanrobincallhere.com	https://americanrobinphotoshere.com
3	Blue Jay	Cyanocitta cristata	Uncommon	https://bluejaycallhere.com	https://bluejayphotoshere.com
4	Northern Cardinal	Cardinalis cardinalis	Uncommon	https://northerncardinalcallhere.com	https://northerncardinalphotoshere.com
5	Red-tailed Hawk	Buteo jamaicensis	Common	https://red-tailedhawkcallhere.com	https://red-tailedhawkphotoshere.com
6	Great Blue Heron	Ardea herodias	Uncommon	https://greatblueheroncallhere.com	https://greatblueheronphotoshere.com
7	American Crow	Corvus brachyrhynchos	Common	https://americancrowcallhere.com	https://americancrowphotoshere.com
8	Common Raven	Corvus corax	Common	https://commonravencallhere.com	https://commonravenphotoshere.com
9	Mourning Dove	Zenaida macroura	Common	https://mourningdovecallhere.com	https://mourningdovephotoshere.com

Table: Birds (part 2 of 3 — measurements and season)

birdID	wingspan	size	matingSeason
1	The average Bald Eagle wingspan is 6.9 feet	The average Bald Eagle is 27.9 inches to 37.8 inches long	Bald Eagle nesting season typically begins in December, and lasts until July. Though their courtship behaviors may begin as early as late Fall, depending on location.
2	The average American Robin wingspan is 12.2 to 15.8 inches	The average American Robin is 7.9 to 11 inches long	American Robin nesting season begins as early as March in warmer regions, with most activity occurring through April until July. Pairs may raise two or three broods per season.
3	The average Blue Jay wingspan is 13.4 to 16.9 inches	The average Blue Jay is 9.8 to 11.8 inches long	Blue Jay nesting season begins mid-March up to July
4	The average Northern Cardinal wingspan is 9.8 to 12.2 inches	The average Northern Cardinal is 8.3 to 9.1 inches long	Northern Cardinal nesting season lasts between March and September

birdID	wingspan	size	matingSeason
5	The average Red-tailed Hawk wingspan is 44.9 to 52.4 inches	The average female Red-tailed Hawk is 19.7 to 25.6 inches long while the males are 17.7 to 22.1 inches long	Red-tailed Hawk nesting season lasts from February to early mid-March
6	The average Great Blue Heron wingspan is 5.5 to 6.6 feet	The average Great Blue Heron is 3.2 to 4.5 feet long	Great Blue Heron nesting season lasts from April to May
7	The average American Crow wingspan is 2.8 to 3.3 feet	The average American Crow is 1.3 to 1.7 feet long	American Crow nesting season lasts from February to May
8	The average Common Raven wingspan is 3.81 to 3.88 feet	The average Common Raven is 1.84 to 2.27 feet long	Common Raven nesting season lasts from February to May
9	The average Mourning Dove wingspan is 1.48 feet	The average Mourning Dove is 9.1 to 13.4 inches long	Mourning Dove nesting season lasts from March to May

Table: Birds (part 3 of 3 — descriptive text)

birdID	identifyingMarks	range	description
1	The Bald Eagle is easily identified by its white capped head and neck, in contrast to the rest of its dark brown coat	Found all across North America, except for the most northern regions and below Mexico	The Bald Eagle got its name from the Middle English word, 'Balde', meaning white-headed (not hairless!) These eagles mainly eat fish, and can be found around bodies of water. Though more often than not, they prefer to steal fish from other fishing animals, humans included.
2	The American Robin can be identified by their dark heads, warm orange underbellies, gray-brown bodies, and white patches under their tails	Found all across North America	American Robins are common all across North America, known for being found year-round anywhere south of Canada. Birds that breed above that dividing line leave for the U.S. when winter approaches. Early signifiers of Spring, they can be seen foraging through lawns during this time.
3	The Blue Jay is white to light grey underneath, and has various shades of blue along its back and crest. Its head is circled with a necklace of black feathers.	Found mainly on the eastern side of the midwestern U.S. and southeastern side of Canada	Blue Jays are intelligent, known to hide away acorns in various locations before winter arrives, and remember most of them during and after winter. The ones they forget help propagate new trees. They have been observed rehidng their acorns if they notice another Blue Jay sees them burying it, preventing the theft of their nutritious treasure.
4	The Northern Cardinal has a reddish beak surrounded by black around its edges. Males are brilliant red all over. Females are pale brown with reddish tinges in their wings, tail, and crest.	They are found mainly on the eastern side of the midwestern U.S. and along the northeast coast of Mexico	Northern Cardinals are very popular birds, being the state bird for seven U.S. states. They also are one of the few songbird species that have females that sing as well. Both males and females can often be seen

birdID	identifyingMarks	range	description
			aggressively attacking their reflections in spring and early summer, when they are the most territorial.
5	The Red-tailed Hawk can be identified by a brown coat above and a pale coat below, with a streaked belly and a dark bar between the shoulder and wrist on the underside of its wings. Its tail is pale below, and a cinnamon-red above.	They are found all across North America, with their range spanning all the way from Canada to the northernmost regions of Nicaragua, and even the Caribbean Islands.	Red-tailed Hawks have screeches that are extremely raspy and shrill. Often when you see eagles and other hawks on television, the call you hear is almost always one of a Red-tailed Hawk. They are also known for swooping courting rituals, and hunting in pairs. They are the most common hawk in North America.
6	The Great Blue Heron can be identified by its distinct blue-gray colour, and wide black stripe over the eye and head. In flight, its wings are two-toned, pale on the forewing and darker on the flight feathers. A subspecies in coastal southern Florida is known to be pure white.	The Great Blue Heron lives all across the U.S., though they can be seen all across Central America during the nonbreeding season and found in southern midwest Canada and northern midwest U.S.A. during the breeding season.	Great Blue Herons, like most Herons, are fishing birds. They are often found wading through the edges of rivers, ponds, and lakes. They often move slowly while peering into the depths before striking at speed at fish.
7	The American Crow can be identified by its all-black color and distinctive hoarse, cawing call. They tend to have fan-shaped tails, as opposed to the diamond-shaped tails of their cousin, the Raven.	The American Crow lives all across the U.S., save for the southwestern deserts, and can be seen as far north as Canada during their breeding season.	American Crows eat mainly earthworms, insects, small animals, seeds, and fruit. However, they aren't picky and eat garbage, carrion, and chicks from other nests. Known widely for their cunning, they have been known to distract other animals to steal food from them.
8	The Common Raven can be identified by its all-black coat. Often larger than crows, they have diamond-shaped tails, and a deeper, croaking call, compared to Crows.	The Common Raven lives on the eastern side of the U.S. Midwest, and all of the area north of Canada. They also live on the coasts of Greenland, and nearly all of Eurasia.	Ravens are known for their superior intelligence, even compared to other members of the Corvid family. They have been solving ever more complex problems given to them by ever more creative scientists.
9	The Mourning Dove can be identified by its plump body, long tail, and relatively small head. They're often brown to tan overall with black spots on their wings, and black-bordered white tips on their tails.	Mourning Doves live all across the U.S. and Mexico, with some territory even being in the southernmost regions of Canada, and the Caribbean islands.	Mourning Doves feed on seeds on the ground, pecking their way through open country or lawns. On the daily, they eat 12 to 20 percent of their body weight, and can drink water that's up to almost half the salinity of the sea.

Table: Nests

nestID	type	location	nestID	type	location	nestID	type	location
1	Scrape	Tree	21	Domed	Brush	41	Globular	Water
2	Scrape	Cliff	22	Domed	Ground	42	Globular	Man-Made Surface
3	Scrape	Brush	23	Domed	Water	43	Cavity	Tree

nestl D	type	location	nestl D	type	location	nestl D	type	location
4	Scrape	Ground	24	Domed	Man-Made Surface	44	Cavity	Cliff
5	Scrape	Water	25	Pendulous	Tree	45	Cavity	Brush
6	Scrape	Man-Made Surface	26	Pendulous	Cliff	46	Cavity	Ground
7	Platform	Tree	27	Pendulous	Brush	47	Cavity	Water
8	Platform	Cliff	28	Pendulous	Ground	48	Cavity	Man-Made Surface
9	Platform	Brush	29	Pendulous	Water	49	Burrow	Tree
10	Platform	Ground	30	Pendulous	Man-Made Surface	50	Burrow	Cliff
11	Platform	Water	31	Pensile	Tree	51	Burrow	Brush
12	Platform	Man-Made Surface	32	Pensile	Cliff	52	Burrow	Ground
13	Cup	Tree	33	Pensile	Brush	53	Burrow	Water
14	Cup	Cliff	34	Pensile	Ground	54	Burrow	Man-Made Surface
15	Cup	Brush	35	Pensile	Water	55	Mound	Tree
16	Cup	Ground	36	Pensile	Man-Made Surface	56	Mound	Cliff
17	Cup	Water	37	Globular	Tree	57	Mound	Brush
18	Cup	Man-Made Surface	38	Globular	Cliff	58	Mound	Ground
19	Domed	Tree	39	Globular	Brush	59	Mound	Water
20	Domed	Cliff	40	Globular	Ground	60	Mound	Man-Made Surface

Table: BirdsNests

birdID	nestID
1	7
1	8
2	14
3	14
4	15
5	7
5	8
6	7
7	7
8	8
9	7
9	10

Table: Birders

birderID	birderName	points
1	Joseph Eidel	13
2	Ayush Baruah	15
3	Leonel Messi	9
4	David Attenborough	61

Table: Sightings

sightingID	birderID	birdID	birdCount	gpsLocation	time
1	1	7	4	NULL	NULL
2	1	9	2	NULL	NULL
3	1	4	2	NULL	NULL
4	1	7	3	NULL	NULL
5	2	7	5	NULL	NULL
6	2	2	1	NULL	NULL
7	2	5	2	NULL	NULL
8	2	1	1	NULL	NULL
9	2	8	2	NULL	NULL
10	2	6	2	NULL	NULL
11	2	9	3	NULL	NULL
12	3	8	5	NULL	NULL
13	3	3	2	NULL	NULL
14	3	9	2	NULL	NULL
15	4	1	3	NULL	NULL
16	4	2	3	NULL	NULL
17	4	3	7	NULL	NULL
18	4	4	13	NULL	NULL
19	4	7	35	NULL	NULL

Table: BirdsList

birderID	birdID	count
1	7	7
1	9	2
1	4	2
2	7	5

birderID	birdID	count
2	2	1
2	5	2
2	1	1
2	8	2
2	6	2
2	9	3
3	8	5
3	3	2
3	9	2
4	1	3
4	2	3
4	3	7
4	4	13
4	7	35

Table: Rewards

rewardID	name	description	threshold
1	On My Way	Spotted your first 10 birds!	10
2	Amateur Birder	Spotted 25 birds!	25
3	True Birder	Spotted 50 birds!	50

Table: BirdersRewards

rewardID	birderID
1	1
1	2
1	4
2	4
3	4

UI Images

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Avidex Home Pag

Developed by Ayush Baruah and Joseph Eidel

Progress Report for Reviewer

Hey reviewers, **ALL CRUD** operations should be functional for **ALL ENTITIES**.

Happy Reviewing!

Avian Compendium: CREATE/READ/UPDATE/DELETE Birds Entity

[illegible]

Add a Bird

Rank:	Select a Rank
Common Name:	
Species:	
Call ID:	
Wingspan:	
Sex:	
Identifying Marks:	
Range:	
Description:	
Photographs List:	
Mating Season:	
Add Bird	

Update a Bird

Bird to Update:	Swain's Thrush	v
Rarity:	Swain's Thrush	v
Common Name:		
Species:		
Catch Date:		
Wingspan:		
Size:		
Identifying Marks:		
Range:		
Description:		
Parasitologic Data:		
Mating Season:		
Update Bird		

- A list of all the birds held within the Avidex servers that users can find and track on the site

Rarities: CREATE/READ/UPDATE/DELETE Rarities Entity

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All Rarities

rarityID	
Common	Delete
Legendary	Delete
Rare	Delete
Uncommon	Delete

Add a Rarity

Rarity Name:

[Add Rarity](#)

Update a Rarity

Rarity to Update:

Rarity Name:

[Update Rarity](#)

- A list of the rarity types for all the birds

Birders: CREATE/READ/UPDATE/DELETE Birders Entity

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Birders

birderID	birderName	points	
1	Joseph Eidel	13	Delete
2	Ayush Baruah	15	Delete
3	Leonel Messi	9	Delete
4	David Attenborough	61	Delete

Add a Birder

Birder Name:

Birder's Points:

[Add Birder](#)

Update a Birder

Birder to Update:

Birder Name:

Birder's Points:

[Update Birder](#)

- A list of all the Birders that have an account

Nests: CREATE/READ/UPDATE/DELETE Nests Entity

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Nests

nestID	type	location	
1	Scrape	Tree	Delete
2	Scrape	Cliff	Delete
3	Scrape	Brush	Delete
4	Scrape	Ground	Delete
5	Scrape	Water	Delete
6	Scrape	Man-Made Surface	Delete
7	Platform	Tree	Delete
8	Platform	Cliff	Delete
9	Platform	Brush	Delete
10	Platform	Ground	Delete
11	Platform	Water	Delete
12	Platform	Man-Made Surface	Delete
13	Cup	Tree	Delete
14	Cup	Cliff	Delete
15	Cup	Brush	Delete
16	Cup	Ground	Delete
17	Cup	Water	Delete

- A list of all the possible nest types (A cross product of the location and type of nest)

Rewards: CREATE/READ/UPDATE/DELETE Rewards Entity

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Rewards

rewardID	name	description	threshold	
1	On My Way	Spotted your first 10 birds!	10	Delete
2	Amateur Birder	Spotted 25 birds!	25	Delete
3	True Birder	Spotted 50 birds!	50	Delete

Add a Reward

Reward Name:

Reward Description:

Reward Threshold:

[Add Reward](#)

Update a Reward

Reward to Update:

Reward Name:

Reward Description:

Reward Threshold:

[Update Reward](#)

- A list of all the rewards the birds can get

Sightings: CREATE/READ/UPDATE/DELETE Sightings Entity

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Sightings

sightingID	birderName	commonName	birdCount	gpsLocation	time	
1	Joseph Eidel	American Crow	4	None	None	Delete
2	Joseph Eidel	Mourning Dove	2	None	None	Delete
3	Joseph Eidel	Northern Cardinal	2	None	None	Delete
4	Joseph Eidel	American Crow	3	None	None	Delete
5	Ayush Baruah	American Crow	5	None	None	Delete
6	Ayush Baruah	American Robin	1	None	None	Delete
7	Ayush Baruah	Red-tailed Hawk	2	None	None	Delete
8	Ayush Baruah	Bald Eagle	1	None	None	Delete
9	Ayush Baruah	Common Raven	2	None	None	Delete
10	Ayush Baruah	Great Blue Heron	2	None	None	Delete
11	Ayush Baruah	Mourning Dove	3	None	None	Delete
12	Leonel Messi	Common Raven	5	None	None	Delete
13	Leonel Messi	Blue Jay	2	None	None	Delete
14	Leonel Messi	Mourning Dove	2	None	None	Delete
15	David Attenborough	Bald Eagle	3	None	None	Delete
16	David Attenborough	American Robin	3	None	None	Delete
17	David Attenborough	Blue Jay	7	None	None	Delete

- A list of all the sightings that a user has made

BirdersRewards: CREATE/READ/UPDATE/DELETE BirdersRewardsEntity

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BirdersRewards

birderID	birderName	rewardID	rewardName	
1	Joseph Eidel	1	On My Way	Delete
2	Ayush Baruah	1	On My Way	Delete
4	David Attenborough	1	On My Way	Delete
4	David Attenborough	2	Amateur Birder	Delete
4	David Attenborough	3	True Birder	Delete

Add a BirderReward

Reward:

Select Reward:

Birder:

Select Birder:

Add BirderReward

Update a BirderReward

Current Birder:

Select current birder:

Current Reward:

Select current reward:

New Birder:

Select new birder:

New Reward:

Select new reward:

Update BirderReward

- An intersection table holding each Birders' Rewards

BirdsList: CREATE/READ/UPDATE/DELETE BirdsList Entity

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Reset

BirdsList

birderID	birderName	birdID	commonName	count	
1	Joseph Eidel	4	Northern Cardinal	2	Delete
1	Joseph Eidel	7	American Crow	7	Delete
1	Joseph Eidel	9	Mourning Dove	2	Delete
2	Ayush Baruah	1	Bald Eagle	1	Delete
2	Ayush Baruah	2	American Robin	1	Delete
2	Ayush Baruah	5	Red-tailed Hawk	2	Delete
2	Ayush Baruah	6	Great Blue Heron	2	Delete
2	Ayush Baruah	7	American Crow	5	Delete
2	Ayush Baruah	8	Common Raven	2	Delete
2	Ayush Baruah	9	Mourning Dove	3	Delete
3	Leonel Messi	3	Blue Jay	2	Delete
3	Leonel Messi	8	Common Raven	5	Delete
3	Leonel Messi	9	Mourning Dove	2	Delete
4	David Attenborough	1	Bald Eagle	3	Delete
4	David Attenborough	2	American Robin	3	Delete
4	David Attenborough	3	Blue Jay	7	Delete
4	David Attenborough	4	Northern Cardinal	13	Delete

- A list of the total sightings of each specific bird a user has seen

BirdsNests: CREATE/READ/UPDATE/DELETE BirdsNests Entity

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Reset

BirdsNests

birdID	commonName	nestID	type	location	
1	Bald Eagle	7	Platform	Tree	Delete
1	Bald Eagle	8	Platform	Cliff	Delete
2	American Robin	14	Cup	Cliff	Delete
3	Blue Jay	14	Cup	Cliff	Delete
4	Northern Cardinal	15	Cup	Brush	Delete
5	Red-tailed Hawk	7	Platform	Tree	Delete
5	Red-tailed Hawk	8	Platform	Cliff	Delete
6	Great Blue Heron	7	Platform	Tree	Delete
7	American Crow	7	Platform	Tree	Delete
8	Common Raven	8	Platform	Cliff	Delete
9	Mourning Dove	7	Platform	Tree	Delete
9	Mourning Dove	10	Platform	Ground	Delete

Add a BirdNest

Bird:

Nest:

Add BirdNest

Update a BirdNest

A list detailing each bird's type of nest

Citations

All About Birds, Cornell Lab, www.allaboutbirds.org/guide/. Accessed 23 July 2026.

Audubon, John James. "Bird Nesting Habits by Species: Fascinating Insights Revealed - Birds." Birdopedia, 13 Nov. 2025, birdopedia.com/bird-nesting-habits-by-species/.

"Bald Eagle Physical Characteristics | American Eagle Foundation." American Eagle Foundation, eagles.org/what-we-do/educate/learn-about-eagles/bald-eagle-physical-characteristics/. Accessed 23 July 2026.

Carver, Erin. U.S. Department of the Interior, U.S. Fish and Wildlife Service, 2024, Birding in the United States: A Demographic and Economic Analysis, <https://www.fws.gov/sites/default/files/documents/2024-11/2022-birding-in-the-us-demographic-and-economic-analysis.pdf>. Accessed 7 July 2026.

Chesser, R. T., et al. "Check-List of North American Birds (Online)." American Ornithological Society, 2025, checklist.americanornithology.org/taxa/.

Thompson, Mya, and Marla Coppelino. "How Nest Designs Protect Eggs and Baby Birds | Bird Academy • the Cornell Lab." Cornell Lab • Bird Academy, Cornell, 6 Mar. 2026, academy.allaboutbirds.org/understand-how-nest-designs-protect-baby-birds/.

Wild-Bird-Watching, www.wild-bird-watching.com/. Accessed 23 July 2026.